3.33 Extensions of Power Series Rings

We record the Weierstrass division lemma for formal power series, used in the analysis of finite extensions of $k((t))$ below.

Remark 3.33.1
*[Che51] (Ch. II, §6) treats the case of (3.33.5) where the ramification index e is prime to $\mathrm{char}(k)$, obtaining the sharper conclusion that t itself is an e-th power in $k((u))$ (a literal Puiseux expansion $k\big(\big(t^{1/e}\big)\big)$), via Hensel's lemma applied to $W^e - v$ for the unit $v := t/u^e$. That argument is unavailable when $\mathrm{char}(k) = p \mid e$: the derivative eW^{e-1} vanishes identically, so $W^e - v$ has no simple root to lift, and t need not be an e-th power at all (an Artin-Schreier extension gives a counterexample). (3.33.5) instead only produces $t = u^e c$ for **some** unit c, obtained by solving $\tau G(\tau) = u^e$ by direct coefficient-stabilisation rather than root extraction; this sidesteps the characteristic obstruction entirely, at the cost of the sharper radical form, which is in any case not needed below.*

Lemma 3.33.2 (Free Basis for a Distinguished Power Series)
Let R be a commutative ring, $m \geq 1$ an integer, $w \in R[[X]]^\star$, and $T := X^m w \in R[[X]]$ (so the coefficients of T below degree m vanish, and its coefficient of X^m is $w(0) \in R^\star$). Then substitution $Y \mapsto T$ (well-defined as T has zero constant term (3.10.5)) is an injective ring homomorphism $R[[Y]] \to R[[X]]$, $f \mapsto f(T)$, and $R[[X]]$ is a free $R[[Y]]$-module of rank m with basis $1, X, \ldots, X^{m-1}$. Explicitly, every $p \in R[[X]]$ is uniquely

$$p = \sum_{i=0}^{m-1} f_i(T) X^i$$

for power series $f_0, \ldots, f_{m-1} \in R[[Y]]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Division by T. Every $p \in R[[X]]$ is uniquely $p = qT + r$ with $q \in R[[X]]$ and $r \in R[X]$ of degree $< m$. For existence, split $p = X^m p' + r$ with r the truncation below degree m and p' the shift, and put $q := p' w^{-1}$ (as $w \in R[[X]]^\star$ (3.10.10)). For uniqueness, if $qT + r = 0$ then $r = -qT \in X^m R[[X]]$ has degree $< m$, so $r = 0$; then $qX^m w = 0$ gives $q = 0$, as w is a unit and X^m is a non-zero-divisor. In particular T is a non-zero-divisor.

Existence. Put $p_0 := p$ and, dividing repeatedly by T, write $p_j = p_{j+1} T + r_j$ with $\deg r_j < m$. Then for every $J \geq 0$

$$p = p_{J+1} T^{J+1} + \sum_{j=0}^{J} r_j T^j, \quad \text{with } p_{J+1} T^{J+1} \in X^{m(J+1)} R[[X]].$$

So $\big(r_j T^j\big)_{j \geq 0}$ is summable (3.10.8) and $p = \sum_{j \geq 0} r_j T^j$. Write $r_j = \sum_{i=0}^{m-1} \gamma_{j,i} X^i$ and set $f_i := \sum_{j \geq 0} \gamma_{j,i} Y^j \in R[[Y]]$. Then

$$\sum_{i=0}^{m-1} f_i(T) X^i = \sum_{i=0}^{m-1} \sum_{j \geq 0} \gamma_{j,i} T^j X^i = \sum_{j \geq 0} r_j T^j = p.$$

Uniqueness. Suppose $\sum_{i=0}^{m-1} f_i(T) X^i = 0$; we show $f_i \in Y^N R[[Y]]$ for all i and all $N \geq 0$, whence every $f_i = 0$. The case $N = 0$ is trivial. Suppose $f_i = Y^N g_i$ for all i. Substitution is a ring homomorphism (3.10.18), so $f_i(T) = T^N g_i(T)$ and $T^N \sum_i g_i(T) X^i = 0$; as T is a non-zero-divisor, $\sum_i g_i(T) X^i = 0$. Write $g_i = c_i + Y h_i$ with $c_i \in R$. Then

$$0 = \sum_{i=0}^{m-1} g_i(T) X^i = \left(\sum_{i=0}^{m-1} h_i(T) X^i \right) T + \sum_{i=0}^{m-1} c_i X^i,$$

which is a division of 0 by T, so uniqueness of division gives $c_i = 0$ for every i. Hence $f_i = Y^{N+1} h_i$ for every i, completing the induction.

Module structure. Substitution $f \mapsto f(T)$ is a ring homomorphism (3.10.18), and it is injective: if $f(T) = 0$ then $f(T) X^0 + \sum_{i=1}^{m-1} 0 \cdot X^i = 0$, so uniqueness (with $p = 0$) forces $f = 0$. Regarding $R[[X]]$ as an $R[[Y]]$-module through it, $f \cdot p = f(T) p$, existence says $1, X, \ldots, X^{m-1}$ generate, and uniqueness says they are linearly independent. ∎

Example 3.33.3
Take $w = 1$, so $T = X^m$. For $p = \sum_{n \geq 0} a_n X^n \in R[[X]]$, writing each $n = mj + i$ with $0 \leq i < m$ gives

$$p = \sum_{i=0}^{m-1} \left(\sum_{j \geq 0} a_{mj+i} X^{mj} \right) X^i, \quad \text{so} \quad f_i(Y) = \sum_{j \geq 0} a_{mj+i} Y^j.$$

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That is, f_i collects the coefficients of p whose exponent is congruent to i modulo m. For $m=2$, f_0 collects the even-index coefficients and f_1 the odd-index ones.

Corollary 3.33.4 (Specialisation Compatibility)
In the setting of (3.33.2), let $\phi : R \to S$ be a homomorphism of commutative rings, extended coefficientwise to $\phi : R[[X]] \to S[[X]]$. Then $\phi(w) \in S[[X]]^\star$, and the decomposition of $\phi(p)$ relative to $\phi(T) = X^m \phi(w)$ is $\phi(p) = \sum_{i=0}^{m-1} \phi(f_i)(\phi(T))X^i$, where $\phi(f_i) \in S[[Y]]$ denotes ϕ applied coefficientwise to f_i.

Proof. If $wv = 1$ then $\phi(w)\phi(v) = 1$, so $\phi(w) \in S[[X]]^\star$. Next, $\phi(f(T)) = \phi(f)(\phi(T))$ for every $f \in R[[Y]]$. Fix $N \geq 0$ and write $f = s + Y^N g$ with $s \in R[Y]$ of degree $< N$ and $g \in R[[Y]]$. Then $f(T) = s(T) + T^N g(T)$ (the tail $\sum_{n \geq N} c_n T^n = T^N \sum_n c_{n+N} T^n$, (3.10.8)), and likewise $\phi(f) = \phi(s) + Y^N \phi(g)$, so $\phi(f)(\phi(T)) = \phi(s)(\phi(T)) + \phi(T)^N \phi(g)(\phi(T))$. Applying the ring homomorphism ϕ to $f(T) = s(T) + T^N g(T)$ gives $\phi(f(T)) = \phi(s)(\phi(T)) + \phi(T)^N \phi(g(T))$, so $\phi(f(T)) - \phi(f)(\phi(T)) \in \phi(T)^N S[[X]] \subseteq X^{mN} S[[X]]$, as $\phi(T) = X^m \phi(w)$. This holds for every N and $\bigcap_N X^N S[[X]] = 0$, so $\phi(f(T)) = \phi(f)(\phi(T))$. Applying ϕ to $p = \sum_i f_i(T) X^i$ therefore gives $\phi(p) = \sum_i \phi(f_i)(\phi(T))X^i$. This exhibits $\phi(p)$ in the form guaranteed unique by (3.33.2) over S, so it is **the** decomposition of $\phi(p)$. ∎

Lemma 3.33.5 (Finite Separable Extensions of $k((t))$ are Power Series Fields)
Let k be algebraically closed, $F := k((t))$, and E/F a finite separable extension of degree n. Then there are $u \in E$, an element $\tau \in uk[[u]]$ with $\tau = u^n c$ for some $c \in k[[u]]^\star$, and a field isomorphism $\Psi : E \xrightarrow{\sim} k((u))$ with $\Psi|_F$ the substitution $t \mapsto \tau$ and $\Psi(u) = u$. In particular $1, u, \ldots, u^{n-1}$ is an F-basis of E.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Step 1: a DVR. Let $\mathcal{O}_F := k[[t]]$, a DVR with fraction field F (3.10.13) (so a Dedekind domain (3.32.3)), and let C be the integral closure of $\mathcal{O}_F$ in E. By (3.24.26) C is a finite $\mathcal{O}_F$-module, and by (3.32.11) it is Dedekind. By lying-over (3.24.17) there is a prime $\mathfrak{p} \subset C$ with $\mathfrak{p} \cap \mathcal{O}_F = (t)$. Put $B := C_{\mathfrak{p}}$, a DVR with fraction field E, and let $e := v_B(t) \geq 1$.

Step 2: residue field and valuation. $C/\mathfrak{p}$ is a domain, finite over $\mathcal{O}_F/(t) = k$, so $k \hookrightarrow C/\mathfrak{p}$ is injective and integral, hence $C/\mathfrak{p}$ is a field (3.24.14), finite over k. As $B = C_{\mathfrak{p}}$, (3.6.43) gives $B/\mathfrak{m}_B \cong \mathrm{Frac}(C/\mathfrak{p}) = C/\mathfrak{p}$ ($C/\mathfrak{p}$ already a field); as k is algebraically closed, $k \to B/\mathfrak{m}_B$ is an isomorphism. Every $a \in \mathcal{O}_F^\times$ is a unit of B, so $v_B(x) = e v_F(x)$ for $x \in F^\times$. Let u be a uniformiser of B, so $t = u^e w$ with $w \in B^\star$.

Step 3: expansions. Every $b \in B$ can be written $b = \sum_{i<e} f_i u^i$ with $f_i \in k[[t]]$. We show by induction on $N \geq 0$ that $b \equiv \sum_{i<e} \left(\sum_{j<N} c_{ij} t^j \right) u^i \pmod{t^N B}$ for some $c_{ij} \in k$. The case $N = 0$ is trivial. For the inductive step, write the current remainder as $t^N y$ with $y \in B$ (inductive hypothesis); since $B/uB = k$, repeated division by u gives $y = \sum_{i<e} d_i u^i + u^e y'$ with $d_i \in k$, $y' \in B$ (there is no need to iterate past degree $e-1$: stop as soon as e terms are extracted). Using $u^e = tw^{-1}$,

$$t^N y = \sum_{i<e} (t^N d_i) u^i + t^{N+1}(w^{-1} y'),$$

with $w^{-1}y' \in B$; so taking $c_{i,N} := d_i$ advances the induction, with new remainder $t^{N+1}(w^{-1}y') \in t^{N+1}B$. Put $f_i := \sum_j c_{ij} t^j \in k[[t]]$. Then $z := b - \sum_i f_i u^i$ satisfies $z \in t^N B$ for every N, so $v_B(z) \geq eN$ for all N and $z = 0$.

Step 4: basis and degree. Every element of E has the form $t^{-N} b$ with $b \in B$, so Step 3 gives $E = \sum_{i<e} F u^i$. These elements are independent: if $\sum_{i<e} f_i u^i = 0$ with $f_i \in F$ not all zero, the valuations $v_B(f_i u^i) = e v_F(f_i) + i$ are distinct modulo e for the nonzero f_i, so the minimum is attained exactly once, a contradiction. Hence $n = e$, and $1, \ldots, u^{e-1}$ is a basis. Moreover $B = \bigoplus_{i<e} k[[t]] u^i$, by the valuation estimate $\min_i(e v_F(f_i) + i) = v_B(b) \geq 0$ applied to $b = \sum_i f_i u^i$ with $f_i \in F$.

Step 5: the minimal polynomial. Write $w^{-1} = \sum_{i<e} g_i u^i$ with $g_i \in k[[t]]$ (Step 3). The constant $g_0(0)$ is the residue of w^{-1}, so $g_0(0) \in k^\star$. From $u^e = tw^{-1}$, u is a root of

$$P(Y) := Y^e - t \sum_{i<e} g_i(t) Y^i \in F[Y].$$

Since $F[u] = E$ has degree e over F, P is the minimal polynomial of u, and $E \cong F[Y]/(P)$ with $Y \mapsto u$.

Step 6: solving for τ. For $s \in uk[[u]]$ put $G(s) := \sum_{i<e} g_i(s) u^i \in k[[u]]$, where $g_i(s)$ is the substitution (well defined as s has zero constant term). Since $G(s) \equiv g_0(0) \pmod{u}$, $G(s)$ is a unit. Define $T(s) := u^e G(s)^{-1} \in u^e k[[u]]$. For $s \equiv s' \pmod{u^N}$ we have $g_i(s) - g_i(s') \in (s - s')k[[u]] \subseteq u^N k[[u]]$, hence $G(s) \equiv G(s') \pmod{u^N}$ and $T(s) \equiv T(s') \pmod{u^{N+e}}$. Starting from $s_0 := 0$ and setting $s_{m+1} := T(s_m)$, the sequence satisfies $s_{m+1} \equiv s_m \pmod{u^{m+e}}$,

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so each coefficient stabilises. Let $\tau \in uk[[u]]$ be the coefficientwise limit. Then $\tau \equiv T(\tau) \pmod{u^N}$ for every N, so $\tau = T(\tau)$, i.e. $\tau G(\tau) = u^e$. Also $\tau = u^e c$ with $c \in k[[u]]^\star$, since $\tau G(\tau) = u^e$ and $G(\tau)$ is a unit.

Step 7: the homomorphism. The substitution $t \mapsto \tau$ defines a ring homomorphism $\sigma : k[[t]] \to k[[u]]$ ((3.10.18), as τ has zero constant term), and since τ is a unit times u^e it extends to a field homomorphism $\sigma : F \to k((u))$. The homomorphism $F[Y] \to k((u))$, $Y \mapsto u$, extending σ, kills P:

$$\sigma(P)(u) = u^e - \tau \sum_{i<e} g_i(\tau) u^i = u^e - \tau G(\tau) = 0.$$

So it factors through $E \cong F[Y]/(P)$, giving a field homomorphism $\Psi : E \to k((u))$ with $\Psi|_F = \sigma$ and $\Psi(u) = u$. It is injective, as E is a field.

Step 8: surjectivity. Since $\tau = u^e c$ with $c \in k[[u]]^\star$, the Weierstrass division lemma (3.33.2) (with $R := k$, $X := u$, $m := e$, $w := c$, so $T = \tau$) gives $k[[u]] = \bigoplus_{i<e} \sigma(k[[t]])u^i$. Multiplying by powers of τ gives $k((u)) = \sum_{i<e} \sigma(F)u^i = \sigma(F)[u]$. But $\operatorname{im}\Psi = \Psi(F[u]) = \sigma(F)[u]$, so Ψ is surjective.

Hence $\Psi : E \xrightarrow{\sim} k((u))$ is an isomorphism of fields over F, with $\Psi(u) = u$ and $\Psi(t) = \tau = u^n c$. ∎

Corollary 3.33.6 (Uniqueness of the Extended Valuation)
In the setting of (3.33.5), let C be the integral closure of $k[[t]]$ in E. Then C has exactly one prime $\mathfrak{p}$ lying over (t), the DVR $C_\mathfrak{p}$ is the unique valuation ring of E containing $k[[t]]$ whose maximal ideal contracts to (t), and its ramification index is $[E : F]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $\mathfrak{q}$ be any prime of C over (t), which exists by lying-over. Steps 2–4 of the proof of (3.33.5) apply verbatim to the DVR $B := C_\mathfrak{q}$ (which has residue field k), and give $e_\mathfrak{q} = v_B(t) = [E : F] = n$.

Now apply the Ramification Formula (3.32.20) to the AKLB situation $k[[t]] \subset C$ over E/F (both Dedekind by (3.32.11), and E/F separable), at the prime (t):

$$\sum_{\mathfrak{q} \mid (t)} e_\mathfrak{q} f_\mathfrak{q} = [C/tC : k] = n.$$

Each $f_\mathfrak{q} = [C/\mathfrak{q} : k]$ is finite, and $C/\mathfrak{q}$ is a field algebraic over k, hence $f_\mathfrak{q} = 1$. So $\sum_\mathfrak{q} e_\mathfrak{q} = n$. By the previous paragraph each term equals n, so there is exactly one prime $\mathfrak{p}$ over (t) and $e_\mathfrak{p} = n$.

For uniqueness of the valuation ring, let $V \subseteq E$ be a valuation ring with $k[[t]] \subseteq V$ and $\mathfrak{m}_V \cap k[[t]] = (t)$. By (3.25.14), $C \subseteq V$. Then $\mathfrak{m}_V \cap C$ is a prime of C over (t), hence equals $\mathfrak{p}$. Every element of $C \setminus \mathfrak{p}$ lies outside $\mathfrak{m}_V$, so is a unit of V; hence $C_\mathfrak{p} \subseteq V$. Since $C_\mathfrak{p}$ is a valuation ring of E and valuation rings are maximal for domination among proper subrings of E (3.25.2), $V = C_\mathfrak{p}$. ∎

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